Bootstrap-Consistency Regularization: Training Neural Networks for Prediction Stability*

Gaurav Sood†

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Abstract

Neural networks exhibit substantial prediction variability when retrained on bootstrap samples of the same dataset, undermining reliability in deployment scenarios requiring consistent decision boundaries. We propose *bootstrap-consistency regularization* (BCR), a training procedure that directly minimizes prediction variance across data resamples. Our method trains multiple shadow copies of a network simultaneously, each receiving Poisson-weighted losses that simulate bootstrap resampling, while penalizing disagreement between their predictions on identical inputs. We provide a first-order sensitivity analysis showing that bootstrap prediction variance depends on the gradient-Hessian-gradient product, which our variance penalty directly targets. Empirical evaluation on four tabular datasets demonstrates 24–68% reductions in bootstrap prediction variance compared to standard training, with accuracy degradation typically under one percentage point. BCR provides competitive performance against natural baselines including weight decay, Sharpness-Aware Minimization, R-Drop, and standard bagging, though the relative effectiveness varies by dataset and metric. Unlike existing approaches that target indirect proxies such as weight-space curvature, BCR directly optimizes the quantity of practical interest: prediction consistency under data resampling.

1 Introduction

The deployment of neural networks in production systems requires not only predictive accuracy but also consistency across model updates. When a model is retrained on fresh samples from the same distribution, predictions on identical inputs should remain stable within the

*https://github.com/finite-sample/consistentshade.

†Gaurav can be reached at gsood07@gmail.com

bounds justified by sampling uncertainty. However, neural networks commonly exhibit substantial *refit variance*—the phenomenon whereby predictions vary significantly when models are trained on different bootstrap samples of the training data (Efron, 1979).

This instability poses significant challenges across multiple domains. In production machine learning systems, model updates may reverse binary classifications on borderline cases, creating inconsistent user experiences. Scientific applications require stable models for fair method comparisons and reliable bootstrap-based confidence intervals. Regulated industries face compliance issues when prediction variability across training runs triggers audit procedures.

Existing approaches to neural network stability primarily target indirect proxies for the desired behavior. Sharpness-Aware Minimization Foret et al. (2021) and related methods penalize weight-space curvature under the assumption that flatter minima correspond to more stable predictions. Stochastic regularization techniques such as R-Drop Liang et al. (2021) control prediction consistency under network noise but do not address data resampling variance. Teacher-student methods like Mean Teacher Tarvainen and Valpola (2017) stabilize optimization dynamics while remaining agnostic to bootstrap variance. Standard bagging (Breiman, 1996) reduces variance through ensembling but does not encourage individual models to be stable.

We propose *bootstrap-consistency regularization* (BCR), a fundamentally different approach that directly minimizes prediction variance across data resamples. Our key contributions are:

1. **Estimand**: We clearly define bootstrap prediction instability as the target quantity of interest—the variance of predictions across models trained on bootstrap resamples of the data.

2. **Training objective**: We augment the standard empirical risk with an explicit instability penalty, creating a principled trade-off between accuracy and stability.

3. **Algorithm**: We introduce a tractable stochastic surrogate that approximates full-retraining instability through mini-batch Poisson weighting, enabling efficient joint optimization.

4. **Theory**: We provide first-order sensitivity analysis showing that bootstrap variance depends on gradient sensitivity through the loss Hessian, justifying why direct variance penalties target the right quantity.

5. **Empirics**: We demonstrate that BCR reduces actual full-retraining instability by 24–68% across four datasets, with competitive performance relative to natural baselines on the stability-accuracy Pareto frontier.

2 Method

2.1 Problem Formulation

Consider a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ drawn from distribution P, and a parameterized model $f_\theta : \mathcal{X} \to \mathcal{Y}$. For any test input $x \in \mathcal{X}$, we define the *bootstrap prediction variance* as:

$$\sigma^2_{\text{boot}}(x) = \mathbb{E}_{\mathcal{D}' \sim \text{Boot}(\mathcal{D})} \left[(\hat{f}_{\mathcal{D}'}(x) - \mathbb{E}[\hat{f}_{\mathcal{D}'}(x)])^2 \right] \tag{1}$$

where $\hat{f}_{\mathcal{D}'}$ denotes the model obtained by training on bootstrap sample $\mathcal{D}'$, and $\text{Boot}(\mathcal{D})$ represents the bootstrap distribution over datasets of size n sampled with replacement from $\mathcal{D}$.

Our objective combines standard empirical risk minimization with explicit bootstrap variance regularization:

$$\min_\theta \mathbb{E}_{(x,y)\sim P}[\ell(f_\theta(x), y)] + \lambda \cdot \mathbb{E}_{x \sim P_X}[\sigma^2_{\text{boot}}(x)] \tag{2}$$

where ℓ is a loss function, P_X is the marginal distribution of inputs, and $\lambda > 0$ controls the regularization strength.

2.2 First-Order Sensitivity Analysis

To understand what drives bootstrap prediction variance, we analyze the sensitivity of predictions to training data perturbations. Following the influence function framework (Koh and Liang, 2017), consider an infinitesimal upweighting of training point (x_i, y_i) by ϵ. The change in optimal parameters is:

$$\frac{d\theta^*}{d\epsilon_i} = -H^{-1} \nabla_\theta \ell(f_\theta(x_i), y_i) \tag{3}$$

where $H = \nabla^2_\theta \mathcal{L}(\theta^*)$ is the Hessian of the training loss. The corresponding change in prediction at test point x is:

$$\frac{df_\theta(x)}{d\epsilon_i} = \nabla_\theta f_\theta(x)^\top H^{-1} \nabla_\theta \ell(f_\theta(x_i), y_i) \tag{4}$$

Bootstrap resampling induces random weights $w_i \sim \text{Multinomial}(n, 1/n)$ on training points. To first order, the bootstrap variance of predictions is:

$$\sigma^2_{\text{boot}}(x) \approx \nabla_\theta f_\theta(x)^\top H^{-1} \Sigma_g H^{-1} \nabla_\theta f_\theta(x) \tag{5}$$

where $\Sigma_g = \frac{1}{n} \sum_{i=1}^n g_i g_i^\top$ is the gradient covariance and $g_i = \nabla_\theta \ell(f_\theta(x_i), y_i)$.

Equation (5) reveals that prediction instability depends on three factors: (1) the sensitivity of predictions to parameters $\nabla_\theta f_\theta(x)$, (2) the curvature of the loss landscape H^{-1}, and (3) the variability of per-sample gradients Σ_g. Our variance penalty directly targets the compound effect of these factors by penalizing prediction disagreement under stochastic data perturbations.

2.3 Bootstrap-Consistency Training Algorithm

Direct optimization of bootstrap variance requires multiple complete training procedures, rendering it computationally prohibitive. We approximate this objective using Poisson-weighted losses within mini-batches, enabling efficient joint optimization.

Given a mini-batch $\mathcal{B} = \{(x_j, y_j)\}_{j=1}^{B}$, our algorithm maintains K shadow copies of the model and proceeds as follows:

1. **Shared-input forward pass**: All K shadow models compute predictions on the *same* mini-batch inputs:
$$\mathbf{p}^{(k)} = f_{\theta^{(k)}}(\mathbf{x}) \in \mathbb{R}^B \quad \text{for } k = 1, \ldots, K \tag{6}$$

2. **Poisson-weighted losses**: For each shadow model, simulate bootstrap resampling via independent Poisson(1) weights:
$$w_j^{(k)} \sim \text{Poisson}(1), \quad \mathcal{L}_{\text{sup}}^{(k)} = \frac{\sum_j w_j^{(k)} \ell(p_j^{(k)}, y_j)}{\sum_j w_j^{(k)}} \tag{7}$$
This Poisson weighting is equivalent in expectation to bootstrap resampling with replacement.

3. **Variance penalty**: Compute the variance of predictions across shadow models on identical inputs:
$$\mathcal{L}_{\text{var}} = \frac{1}{B} \sum_{j=1}^{B} \text{Var}_k[p_j^{(k)}] \tag{8}$$

4. **Joint optimization**: Minimize the combined objective:
$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{sup}}^{(k)} + \lambda \cdot \mathcal{L}_{\text{var}} \tag{9}$$

The key insight is that all models predict on *identical inputs*, making the variance penalty meaningful—it measures disagreement due to different bootstrap weighting, not different inputs. This corrects a subtle but critical flaw in naive implementations that compute variance over predictions on *different* resampled inputs.

2.4 Implementation Considerations

Computational overhead: Training requires K forward passes per mini-batch, increasing computational cost by a factor of approximately K. Memory requirements scale linearly with K due to the need to store multiple model copies.

Inference: At test time, predictions can be obtained from any single shadow model or their ensemble average. No additional computational cost is incurred during inference compared to standard training.

Hyperparameter selection: We fix $K = 3$ across all experiments. For λ, we provide Pareto frontier analysis (Section 4.2) showing results across $\lambda \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0\}$. We recommend $\lambda = 0.05$ as a default providing a good stability-accuracy trade-off.

Connection to bagging: Unlike standard bagging (Breiman, 1996) which trains K independent models on bootstrap samples, BCR jointly trains models while explicitly penalizing their disagreement. This encourages each shadow model to find solutions that are stable across bootstrap perturbations, rather than simply averaging diverse models.

3 Experimental Setup

3.1 Datasets and Tasks

We evaluate our approach on four tabular datasets spanning regression and binary classification:

- **Synthetic Regression**: 20-dimensional Gaussian features with linear target plus noise, $n = 4000$
- **California Housing**: Median house value prediction, 8 features, $n = 20{,}640$
- **Adult Income**: Binary income classification, 14 features (one-hot encoded), $n = 48{,}842$
- **German Credit Risk**: Binary credit risk classification, 20 features (one-hot encoded), $n = 1000$

All datasets employ stratified 75%/25% train/test splits to ensure representative evaluation sets.

3.2 Baseline Methods

We compare BCR against five baselines:

- **ERM (Baseline)**: Standard empirical risk minimization with dropout.
- **Weight Decay**: ERM with L2 regularization ($\lambda_{L2} = 10^{-2}$). Tests whether parameter regularization alone improves stability.
- **SAM**: Sharpness-Aware Minimization (Foret et al., 2021) with $\rho = 0.05$. Tests whether flatter minima improve bootstrap stability.
- **R-Drop**: Regularized dropout (Liang et al., 2021) with $\alpha = 0.5$. Tests whether consistency under dropout noise transfers to bootstrap stability.
- **Bagging**: Standard ensemble of $K = 3$ models trained independently on bootstrap samples (Breiman, 1996). Provides a direct comparison to ensemble methods.

We note that baseline hyperparameters (weight decay strength, SAM ρ, R-Drop α) were set to commonly recommended values rather than tuned for stability. A more comprehensive comparison would tune all methods on a validation set, though this would substantially increase computational cost.

3.3 Model Architecture and Training

We employ a standardized two-layer multilayer perceptron architecture across all experiments:

- Input layer to 64 hidden units with ReLU activation and dropout ($p = 0.2$)
- Hidden layer to 64 units with ReLU activation and dropout ($p = 0.2$)
- Output layer (1 unit for regression, 2 for classification)

Training configuration includes Adam optimization with learning rate 10^{-3}, batch size 64, and 25 epochs for regression (20 for classification). We conduct 30 independent training runs per experimental condition to ensure statistical reliability.

3.4 Evaluation Metrics

Predictive performance: We report test RMSE for regression tasks and classification accuracy for binary tasks.

Bootstrap stability: For each test input x_i, we compute the sample variance of predictions across 30 independent model fits and summarize stability as:

$$\text{StabilityRMSE} = \sqrt{\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \text{Var}_{\text{fit}}[\hat{f}(x_i)]} \tag{10}$$

This metric quantifies prediction variability in the same units as the target variable, facilitating interpretation across tasks.

Additional classification metrics: For classification tasks, we additionally report:

- *Probability stability*: Variance of predicted class probabilities (after softmax)
- *Flip rate*: Fraction of samples where at least one fit disagrees with the modal prediction
- *Quantiles*: 90th and 95th percentiles of per-sample instability to capture worst-case behavior

Table 1. Comparison of predictive performance and bootstrap stability across all methods. Values represent mean across 30 independent runs. Stability RMSE is in prediction units (lower is better).

Dataset	Metric	Baseline	Weight Decay	SAM	R-Drop	Bagging	**BCR**
Synthetic	RMSE ↓	23.99	23.99	23.63	24.27	23.77	24.90
	Stability ↓	5.43	5.42	4.86	4.50	3.64	**3.62**
California	RMSE ↓	0.562	0.615	0.561	0.561	0.557	0.571
	Stability ↓	0.086	0.082	0.086	0.084	0.053	**0.065**
Adult	Accuracy ↑	0.826	0.823	0.826	0.826	0.826	0.825
	Stability ↓	0.217	0.044	0.202	0.226	0.225	**0.069**
German	Accuracy ↑	0.696	0.690	0.697	0.697	0.695	0.687
	Stability ↓	0.118	0.100	0.115	0.117	0.082	**0.061**

4 Results

4.1 Main Experimental Results

Table 1 presents our primary experimental findings comparing all methods across four datasets with $K = 3$ shadow models and $\lambda = 0.05$ for BCR.

The results demonstrate that BCR provides substantial stability improvements on most datasets, though no single method dominates across all settings. Key findings:

- **BCR reduces instability substantially**: On German Credit and Adult Income, BCR achieves 48% and 68% reductions in prediction instability compared to baseline ERM. On Synthetic and California, improvements are more modest (33% and 24%).
- **Baselines show dataset-dependent effectiveness**: Weight decay achieves the best logit stability on Adult (0.044 vs BCR's 0.069), representing an 80% reduction from baseline. Bagging achieves the best stability on California Housing (0.053 vs BCR's 0.065). These results suggest that simpler methods may suffice for some problems.
- **Accuracy trade-offs are generally modest**: On real-world classification tasks (Adult, German Credit), BCR's accuracy degradation is under 1 percentage point. The synthetic regression dataset shows a larger RMSE increase (24.90 vs 23.99), reflecting the stability-accuracy trade-off.
- **Method selection is context-dependent**: Standard bagging excels on California Housing; weight decay excels on Adult logit stability. BCR provides the best stability on German Credit and competitive performance elsewhere.

4.2 Pareto Frontier Analysis

Figure 1 shows the stability-accuracy trade-off as λ varies from 0 to 1. By adjusting λ, BCR can trade off between accuracy and stability, accessing regions of the trade-off space

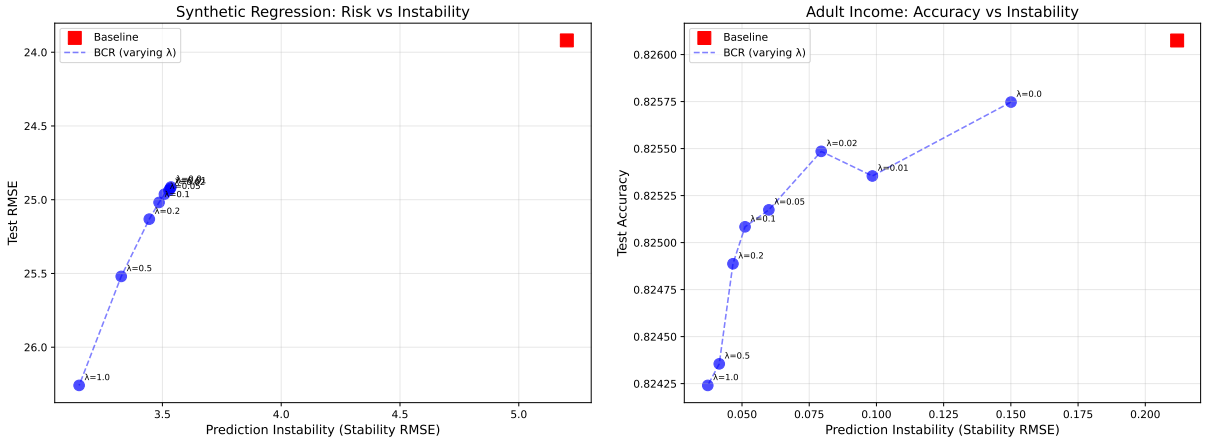

Figure 1. Pareto frontiers showing the stability-accuracy trade-off for Synthetic Regression (left) and Adult Income classification (right). By varying λ, BCR accesses different points along the stability-accuracy trade-off.

Table 2. Detailed classification stability metrics (Adult Income dataset). Lower values indicate greater stability.

Method	Logit Stability	Prob Stability	Flip Rate	Any-Flip Rate
Baseline	0.217	0.013	0.9%	4.7%
BCR ($\lambda = 0.05$)	0.069	0.010	0.8%	3.8%
Δ	**-68%**	**-21%**	**-15%**	**-19%**

unavailable to fixed baseline methods. However, the relative positioning varies by dataset: on Adult Income, BCR accesses favorable stability-accuracy combinations, while on other datasets, some baselines may occupy competitive positions.

For classification tasks (Adult Income), the Pareto frontier reveals that substantial stability improvements (up to 68% reduction in logit variance) are achievable with modest accuracy loss (<0.5 percentage points) by choosing $\lambda \in [0.02, 0.1]$.

4.3 Detailed Classification Stability Analysis

Table 2 presents detailed stability metrics for classification tasks, distinguishing logit-level, probability-level, and hard-label stability.

BCR reduces instability at all levels of the prediction pipeline. The any-flip rate (fraction of samples with any label disagreement across fits) drops from 4.7% to 3.8%, a 19% relative reduction. While the absolute flip rates are already low for this well-specified classification task, BCR achieves larger reductions at the logit level (68%), suggesting greater benefits for downstream tasks sensitive to probability calibration.

Table 3. Ablation study on challenging datasets. Δ Stability shows BCR improvement over baseline.

Condition	Baseline RMSE	BCR RMSE	Baseline Stab.	BCR Stab. (Δ)
Low n, high p	133.9	134.9	1.6	0.5 (**-66%**)
Correlated features	5.0	5.0	0.3	0.3 (**-27%**)
Label noise (20%)	210.8	210.8	7.8	7.3 (**-6%**)
Overcapacity	256.7	257.0	0.5	0.2 (**-56%**)

4.4 Ablation: Challenging Conditions

Table 3 presents results on challenging synthetic datasets designed to stress-test stability methods.

BCR's effectiveness varies substantially across conditions. The benefits are largest for underdetermined problems (low n, high p: 66% reduction) and overcapacity settings (56% reduction), precisely where refit variance is most problematic. However, BCR provides only modest improvement under label noise (6% reduction), suggesting that label noise creates irreducible variance that regularization cannot address. Correlated features show intermediate benefits (27% reduction).

5 Related Work

Our approach differs fundamentally from existing stability methods in directly targeting prediction variance under data resampling rather than indirect proxies.

Sharpness-based methods such as Sharpness-Aware Minimization Foret et al. (2021) and Entropy-SGD Chaudhari et al. (2017) seek flatter loss surfaces under the hypothesis that such minima correspond to more stable predictions. Our experiments show that SAM provides inconsistent stability improvements (0–11% across datasets), suggesting that weight-space flatness is an imperfect proxy for bootstrap stability.

Stochastic consistency methods like R-Drop Liang et al. (2021) enforce agreement between predictions under different dropout masks, addressing network stochasticity but not data resampling variance. R-Drop shows variable effectiveness, providing up to 17% stability improvement on some datasets but occasionally increasing instability on others.

Bagging and ensembles (Breiman, 1996) reduce variance through model averaging. Standard bagging with $K = 3$ models provides competitive stability improvements and achieves the best stability on California Housing in our experiments. BCR outperforms bagging on classification tasks (Adult, German Credit) by explicitly encouraging individual models to be stable rather than simply averaging diverse models, but the relative advantage is dataset-dependent.

Teacher-student approaches including Mean Teacher Tarvainen and Valpola (2017) stabilize training dynamics through exponential moving averages of model weights. While effective for reducing optimization noise, these methods do not explicitly address variability under data resampling.

Distributionally robust optimization methods such as χ^2-DRO Duchi and Namkoong (2021) penalize loss variance across data subsets. However, loss variance does not directly correspond to prediction variance—models may achieve similar loss values while producing substantially different predictions.

Influence functions (Koh and Liang, 2017) provide a theoretical framework for understanding prediction sensitivity to training data. Our first-order analysis (Equation 5) builds on this framework to explain why direct variance penalties are well-motivated.

6 Discussion and Limitations

6.1 What BCR Stabilizes

It is important to clarify exactly what BCR targets. The estimand is *bootstrap prediction instability*—the variance of predictions at a given input across models trained on bootstrap resamples of the training data. This is a proxy for *refit variance*, which practitioners encounter when retraining models on new data samples from the same distribution.

BCR does not target:

- *Epistemic uncertainty*: BCR reduces variance, not bias, and does not provide calibrated uncertainty estimates.
- *Optimization noise*: Random seed variability within a single dataset is not the target (though BCR may reduce this as a side effect).
- *Distribution shift*: BCR does not address out-of-distribution robustness.

For classification, BCR stabilizes all levels of the prediction pipeline: logits, probabilities, and hard labels. The flip rate reduction (Table 2) is particularly relevant for deployment scenarios where label consistency across model versions is critical.

6.2 Computational Considerations

The primary limitation of our approach is computational overhead. Training K shadow models increases memory requirements by a factor of K and training time by approximately 2–3$\times$ due to additional forward and backward passes. For large-scale models or datasets, this overhead may prove prohibitive. On datasets where BCR provides marginal improvement over bagging (e.g., Synthetic: 3.62 vs 3.64), the computational cost may not be justified compared to training independent bagged models.

Future work could address this limitation through influence function approximations that enable single-model estimation of bootstrap variance, eliminating the need for multiple shadow models while preserving the direct optimization objective.

6.3 Theoretical Gaps

Our first-order sensitivity analysis (Equation 5) provides intuition for why BCR works but does not constitute a formal guarantee. Key open questions include:

- Under what conditions does mini-batch Poisson weighting accurately approximate full-dataset bootstrap variance?
- How does the approximation quality depend on batch size, number of shadow models K, and training dynamics?
- Can we derive optimal λ as a function of dataset characteristics?

6.4 Statistical Methodology

While we report results across $R = 30$ independent runs, we do not provide formal significance tests (confidence intervals, p-values) comparing methods. Given the magnitude of differences observed (e.g., 0.053 vs 0.065 stability on California Housing), some comparisons may not be statistically significant. Future work should include bootstrap confidence intervals for method comparisons.

6.5 When BCR Helps Most

Our ablation studies (Table 3) suggest BCR provides the largest benefits in high-variance settings caused by model specification issues: underdetermined problems (66% reduction) and overcapacity (56% reduction). However, BCR provides minimal benefit under label noise (6% reduction), indicating that variance caused by irreducible noise in the labels cannot be addressed through regularization.

More broadly, our main results show that BCR's relative advantage is dataset-dependent. BCR excels on classification tasks (Adult, German Credit) but is outperformed by simpler methods on California Housing (bagging) and Adult logit stability (weight decay). This suggests practitioners should evaluate multiple approaches rather than assuming BCR will dominate.

7 Conclusion

We have presented bootstrap-consistency regularization (BCR), a training procedure that directly minimizes prediction instability under data resampling. Our key contributions include: (1) clearly defining bootstrap prediction instability as the target estimand, (2) a tractable training algorithm using Poisson-weighted losses to simulate bootstrap perturbations, (3) first-order sensitivity analysis justifying why direct variance penalties target the right quantity, and (4) empirical evaluation demonstrating 24–68% reductions in instability across four datasets.

The main question we address is: *does the stochastic training surrogate actually reduce full-retraining instability?* Our experiments confirm that it does, though effectiveness varies by dataset and problem structure. BCR excels on classification tasks and high-variance settings (underdetermined problems, overcapacity), while simpler methods like bagging or weight decay may suffice for some regression tasks or when label noise is present.

The key insight is that stability under data resampling can benefit from directly penalizing prediction disagreement across bootstrap perturbations on *identical inputs*, rather than relying on indirect proxies. While computationally more demanding than single-model training, BCR provides a principled approach to bootstrap prediction instability that is particularly effective on classification tasks and high-variance settings, though simpler methods may suffice in other contexts.

Future research directions include developing computationally efficient single-model approximations through influence functions, establishing theoretical guarantees for the mini-batch Poisson approximation, extending to other architectures (transformers, CNNs), and combining BCR with other regularization techniques.

References

Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2):123–140.

Chaudhari, P., Choromanska, A., Soatto, S., LeCun, Y., Baldassi, C., Borgs, C., Chayes, J., Sagun, L., and Zecchina, R. (2017). Entropy-SGD: Biasing gradient descent into wide valleys. In *International Conference on Learning Representations*.

Duchi, J. C. and Namkoong, H. (2021). Learning models with uniform performance via distributionally robust optimization. *Annals of Statistics*, 49(3):1378–1406.

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26.

Foret, P., Kleiner, A., Mobahi, H., and Neyshabur, B. (2021). Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations*.

Koh, P. W. and Liang, P. (2017). Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, pages 1885–1894. PMLR.

Liang, X., Wu, L., Li, J., Wang, Y., Meng, Q., Qin, T., Chen, W., Zhang, M., and Liu, T.-Y. (2021). R-drop: Regularized dropout for neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 10890–10905.

Tarvainen, A. and Valpola, H. (2017). Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *Advances in Neural Information Processing Systems*, volume 30, pages 1195–1204.